



User Manual



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1 Product Overview

1.1 Product Introduction

The P31-U series is a remote distributed IO host module that supports the Profinet protocol and is used to map IO data to Profinet. It uses a GSD file for integration, and the PLC reads and writes the IO port without code. It supports a variety of Siemens PLC models including Siemens S7-200smart, S7-300, 1S7-200, S7-1500 and S7-1500. It has two network ports, supports switch functions, and integrates Profinet to Modbus RTU gateway functions.

The design adopts a modular splicing structure, which allows users to flexibly expand according to actual needs. When the existing configuration is not sufficient to meet specific application requirements, users can easily add matching IO expansion modules without replacing the entire system, effectively saving costs and simplifying the on-site deployment process.

This product allows the connection of up to 16 IO expansion modules (including the host itself) and is built in accordance with the EMC Level 3 protection standard. The IO input and output, RS485 and power supply are fully isolated, providing excellent electromagnetic compatibility, outstanding performance and high reliability, and complies with CE certification standards.



1.2 Features

- Adopts standard Profinet protocol communication, and can be networked with PLC, configuration, host computer, etc.
- Dual industrial Ethernet ports 10/100M adaptive, built-in switch function
- Support Profinet to Modbus RTU gateway function
- Modular splicing structure, small size and compact structure
- Support differential analog input, resolution 16 bits, accuracy within 1‰
- Millis-level response, internal bus speed up to 1ms
- A maximum of 16 IO expansion modules can be connected (including the host itself)
- The maximum number of IOs supported is 256 bits
- Any combination of switch input and output, analog input and output, etc.
- Industrial-grade hardware design, with multiple protections such as static electricity, lightning surge, fast pulse group , and power reverse connection protection
- With IO input and output, RS485 isolation, power supply isolation
- Support positioning hole and guide rail installation

- Meets CE certification standards

1.3 Product Model List

Product Model	Product Specifications	Switch input DI	Switch output DO	Analog input AI	Analog output AO
P31-AFAX4440G-U	4DI+4DO+4AI	4 (NPN, PNP)	4 (Relay)	4 (Differential current)	—
P31-AXAX8080G-U	8DI+8DO	8 (NPN, PNP)	8 (Relay)	—	—

2 Technical indicators

2.1 Specifications

Profinet Distributed IO Master		
category	name	parameter
Power supply	Operating voltage	DC 9 ~ 36 V ; with reverse connection protection
	Power indicator	PWR red LED indicator
	Power consumption	2~4W
Network port parameters	Interface Type	RJ45 *2 (dual network ports) , RS485
	Communication bandwidth	10/100M adaptive
	Communication Protocol	Standard Profinet protocol (RT)
	Internal bus	Up to 1ms
DI Input	Input Type	NPN, PNP
	Input range	DC 12 ~ 24V
	Isolation method	Each channel is optically isolated
	Input Impedance	7.2kΩ
	Collection frequency	1 000 Hz
	Counting frequency	< 500 Hz
	Filter time	Default time 6ms
	Input instructions	DI green LED indicator
AI Input	Acquisition characteristics	Differential input
	Input Type	Differential current: 0-20mA, 4-20mA, ±20mA
	AI Resolution	16 bits (differential)
	AI Precision	1%(difference)
	Collection frequency	Differential max. 70 Hz
	Input instructions	AI green LED indicator
DO Output	DO output type	Type A relay (normally open)
	DO output mode	Level output , pulse output
	Relay contact capacity	5A 30VDC, 5A 250VAC (the total current of the same COM common terminal supports a maximum of 8A)
	Output indication	DO green LED indicator

Other	Product size	110mm * 40mm * 100mm (length * width * height)
	Working temperature and humidity	-40 ~ +85°C, 5% ~ 95%RH (no condensation)
	Storage temperature and humidity	-40 ~ +105°C, 5% ~ 95%RH (no condensation)
	Installation	Positioning holes and guide rail installation

2.2 Dimensions



2.3 Ports, buttons and LED indicators

2.3.1 P31-AFAX4440G-U

P31-AFAX4440G-U Port and Button Description:		
Silkscreen	name	illustrate
DI1	DI1 switch input	DI1 switch input interface, used with COM A
DI2	DI2 switch input	DI2 switch input interface, used with COM A
COM A	DI switch input common terminal	DI1-DI2 share COM A common terminal
DI3	DI3 switch input	DI3 switch input interface, used with COM B
DI4	DI4 switch input	DI4 switch input interface, used with COM B
COM B	DI switch input common terminal	DI3-DI4 share COM B common terminal
DO1	DO1 switch output	DO1 switch output interface, used with COMC
DO2	DO2 switch output	DO2 switch output interface, used with COMC
DO3	DO3 switch output	DO3 switch output interface, used with COMC
DO4	DO4 switch output	DO4 switch output interface, used with COMC
COM C	COM port of DO	DO1-DO 4 share COMC for use
AI1+	AI1 analog input +	AI1 analog input + interface, used with AI1-
AI1-	AI1 analog input -	AI1 analog input-interface, used with AI1+
AI2+	AI2 analog input +	AI2 analog input + interface, used with AI2-
AI2-	AI2 analog input -	AI2 analog input-interface, used with AI2+
AI3+	AI3 analog input +	AI3 analog input + interface, used with AI3-
AI3-	AI3 analog input -	AI3 analog input-interface, used with AI3+
AI4+	AI4 analog input +	AI4 analog input + interface, used with AI4-
AI4-	AI4 analog input -	AI4 analog input-interface, used with AI4+
Reload	Factory Reset/Auto Negotiation	Long press for 5-10 seconds to restore to factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	Negative pole of power supply	DC (9-36V) power negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power positive terminal interface
LAN1	Network port	Standard RJ45 network cable interface
LAN 2	Network port	Standard RJ45 network cable interface

P31- AFAX4440G-U indicator light description:

Silkscreen	name	illustrate
RUN	Equipment operating status indicator	Red LED light; On: After the connection to the master station is initialized; Off: The device is running abnormally; 1Hz flashing: Not connected to the master station and initialization is not completed; 4Hz flashing: The configuration software actively searches and queries for the device to flash; 20Hz flashing: The gateway serial port sends and receives data;
BUS	Device bus status indicator	Yellow LED light; On: The internal bus of the device is running normally; Off: The internal bus of the device is running completely abnormally; Flashing: The internal bus of the device may be abnormal.
STA	Bus communication indicator	Blue LED light; flashing: indicates that there is data exchange between the host and the expansion module; off: indicates that there is no data exchange between the host and the expansion module
DI1	DI1 input indicator	Green LED light; on: DI1 valid input; off: DI1 invalid input
DI2	DI2 input indicator	Green LED light; on: DI2 valid input; off: DI2 invalid input
DI3	DI3 input indicator	Green LED light; on: DI3 valid input; off: DI3 invalid input
DI4	DI4 input indicator	Green LED light; on: DI4 valid input; off: DI4 invalid input
DO1	DO1 output indicator	Green LED light; on: DO1 relay closed; off: DO1 relay open
DO2	DO2 output indicator	Green LED light; on: DO2 relay closed; off: DO2 relay open
DO3	DO3 output indicator	Green LED light; on: DO3 relay closed; off: DO3 relay open
DO4	DO4 output indicator	Green LED light; on: DO4 relay closed; off: DO4 relay open
AI1	AI1 input indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: No effective access; Fast flashing: Exceeds the range by 10%
AI2	AI2 input indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: No effective access; Fast flashing: Exceeds the range by 10%
AI3	AI3 input indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: No effective access; Fast flashing: Exceeds the range by 10%
AI4	AI4 input indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: No effective access; Fast flashing: Exceeds the range by 10%

2.3.2 P31-AXAX8080G-U

P31-AXAX8080G-U Port and button description:		
Silkscreen	name	illustrate
DO1	DI1 switch input	DI1 switch input interface, used with COMA
DI2	DI2 switch input	DI2 switch input interface, used with COMA
DI3	DI3 switch input	DI3 switch input interface, used with COMA
DI4	DI4 switch input	DI4 switch input interface, used with COMA
COMA	DI switch input common terminal	DI1-DI4 share COMA common terminal
DI5	DI5 switch input	DI5 switch input interface, used with COMB
DI6	DI6 switch input	DI6 switch input interface, used with COMB
DI7	DI7 switch input	DI7 switch input interface, used with COMB
DI8	DI8 switch input	DI8 switch input interface, used with COMB
COMB	DI switch input common terminal	DI5-DI8 share COMB common terminal
DO1	DO1 switch output	DO1 switch output interface, used with COMC
DO2	DO2 switch output	DO2 switch output interface, used with COMC
DO3	DO3 switch output	DO3 switch output interface, used with COMC
DO4	DO4 switch output	DO4 switch output interface, used with COMC
DO5	DO5 switch output	DO5 switch output interface, used with COMC
DO6	DO6 switch output	DO6 switch output interface, used with COMC
DO7	DO7 switch output	DO7 switch output interface, used with COMC
DO8	DO8 switch output	DO8 switch output interface, used with COMC
COMC	COM port of DO	DO1-DO8 share COMC for use
Reload	Factory Reset/Auto Negotiation	Long press for 5-10 seconds to restore to factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	Negative pole of power supply	DC (9-36V) power negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power positive terminal interface
LAN1	Network port	Standard RJ45 network cable interface
LAN 2	Network port	Standard RJ45 network cable interface

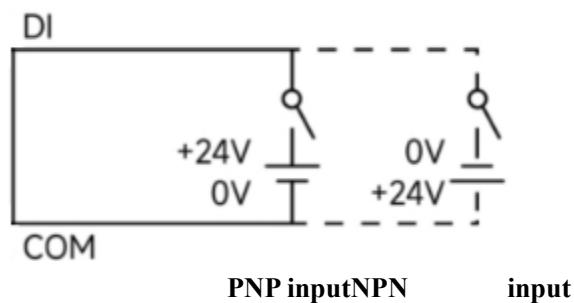
P31- AXAX8080G-U indicator light description:

Silkscreen	name	illustrate
RUN	Equipment operating status indicator	Red LED light; On: After the connection to the master station is initialized; Off: The device is running abnormally; 1Hz flashing: Not connected to the master station and initialization is not completed; 4Hz flashing: The configuration software actively searches and queries for the device to flash; 20Hz flashing: The gateway serial port sends and receives data;
BUS	Device bus status indicator	Yellow LED light; On: The internal bus of the device is running normally; Off: The internal bus of the device is running completely abnormally; Flashing: The internal bus of the device may be abnormal.
STA	Bus communication indicator	Blue LED light; flashing: indicates that there is data exchange between the host and the expansion module; off: indicates that there is no data exchange between the host and the expansion module
DI1	DI1 input indicator	Green LED light; on: DI1 valid input; off: DI1 invalid input
DI2	DI2 input indicator	Green LED light; on: DI2 valid input; off: DI2 invalid input
DI3	DI3 input indicator	Green LED light; on: DI3 valid input; off: DI3 invalid input
DI4	DI4 input indicator	Green LED light; on: DI4 valid input; off: DI4 invalid input
DI5	DI5 input indicator	Green LED light; on: DI5 valid input; off: DI5 invalid input
DI6	DI6 input indicator	Green LED light; on: DI6 valid input; off: DI6 invalid input
DI7	DI7 input indicator	Green LED light; on: DI7 valid input; off: DI7 invalid input
DI8	DI8 input indicator	Green LED light; on: DI8 valid input; off: DI8 invalid input
DO1	DO1 output indicator	Green LED light; on: DO1 relay closed; off: DO1 relay open
DO2	DO2 output indicator	Green LED light; on: DO2 relay closed; off: DO2 relay open
DO3	DO3 output indicator	Green LED light; on: DO3 relay closed; off: DO3 relay disconnected
DO4	DO4 output indicator	Green LED light; on: DO4 relay closed; off: DO4 relay disconnected
DO5	DO5 output indicator	Green LED light; on: DO5 relay closed; off: DO5 relay open
DO6	DO6 output indicator	Green LED light; on: DO6 relay closed; off: DO6 relay disconnected
DO7	DO7 output indicator	Green LED light; on: DO7 relay closed; off: DO7 relay open
DO8	DO8 output indicator	Green LED light; on: DO8 relay closed; off: DO8 relay disconnected

3 Wiring Instructions

3.1 Device Connection

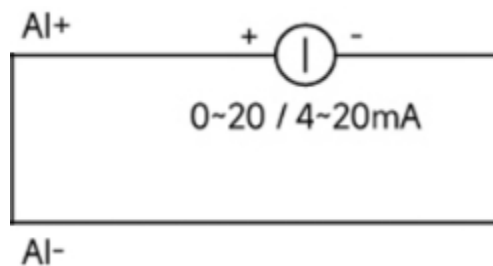
3.1.1 DI Connection



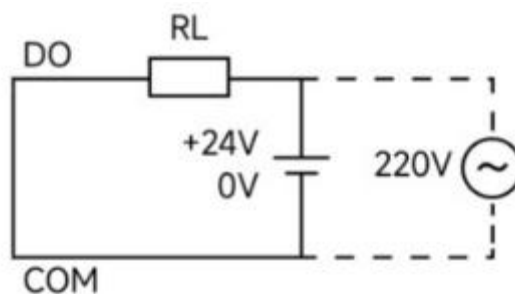
Note: DI is NPN or PNP active input, and the voltage range only supports 10V to 28V .

3.1.2 AI Connection

Differential analog current acquisition:



3.1.3 DO Connection



Note: 1. A single relay supports a maximum of 5A.

2. The maximum total current of each group (same COM common terminal) supports 8A.

4 Tutorial on using Botu to load devices

4.1 Preparation before connection

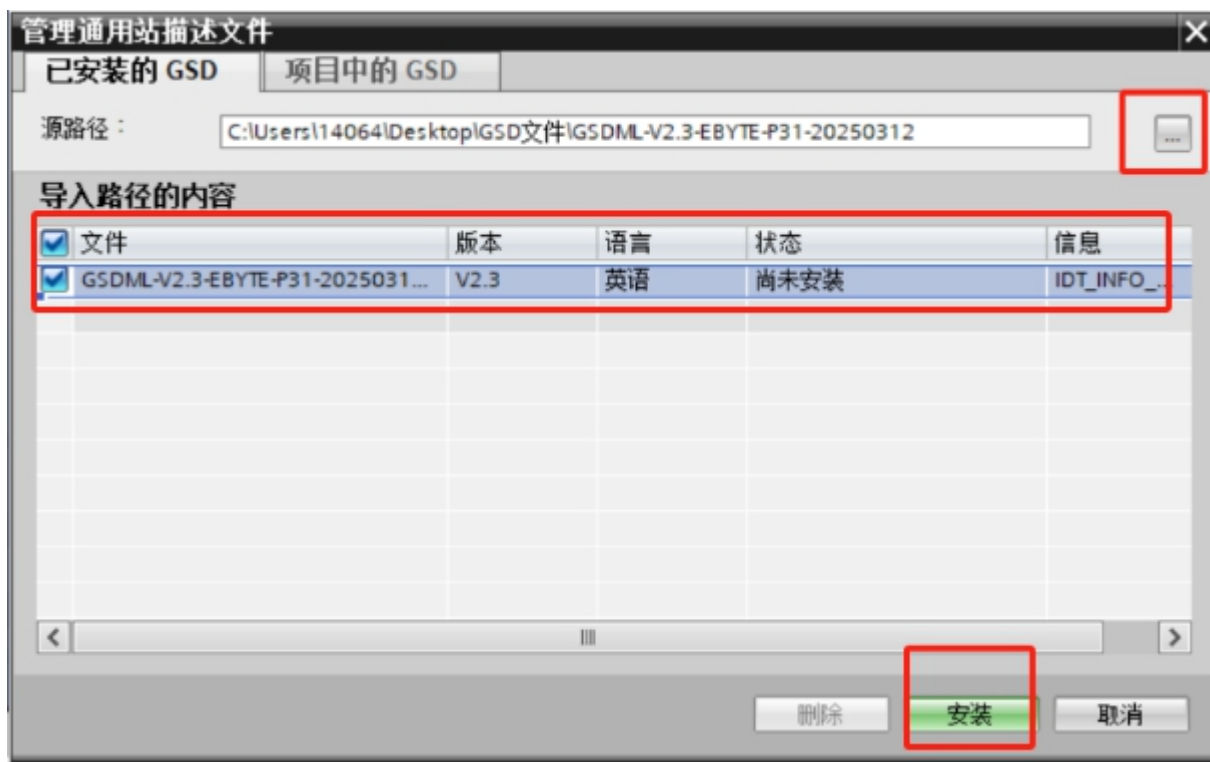
1. Prepare the required XML files as follows:
GSDML-V2.3-EBYTE-P31-20250312.xml

4.2 Add GSDML file to Botu

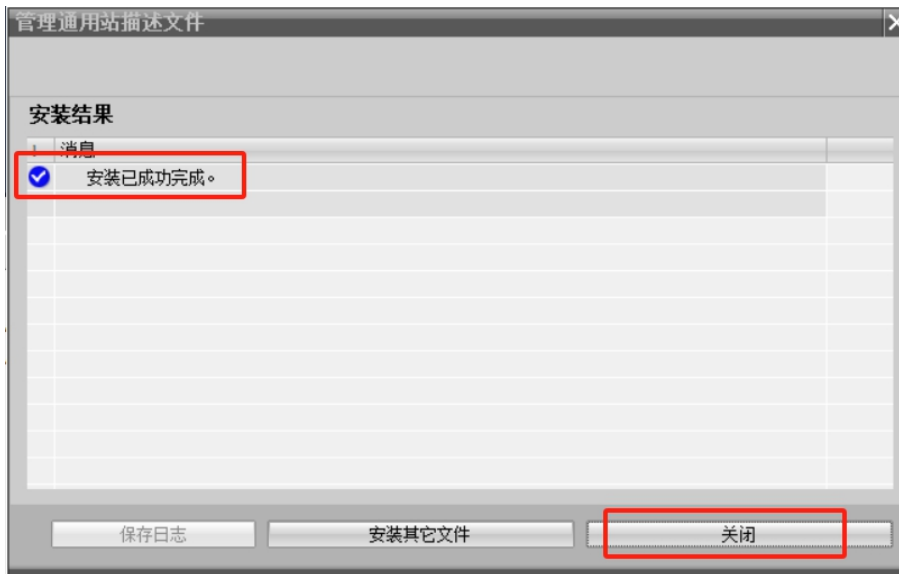
1. Open Botu, click Options, and manage general site files (gsd)



2. Add the GSDML file, find the folder where the GSD is stored, select the folder, and then the software will scan the GSD file by itself, then check the GSD file, and finally click Install.



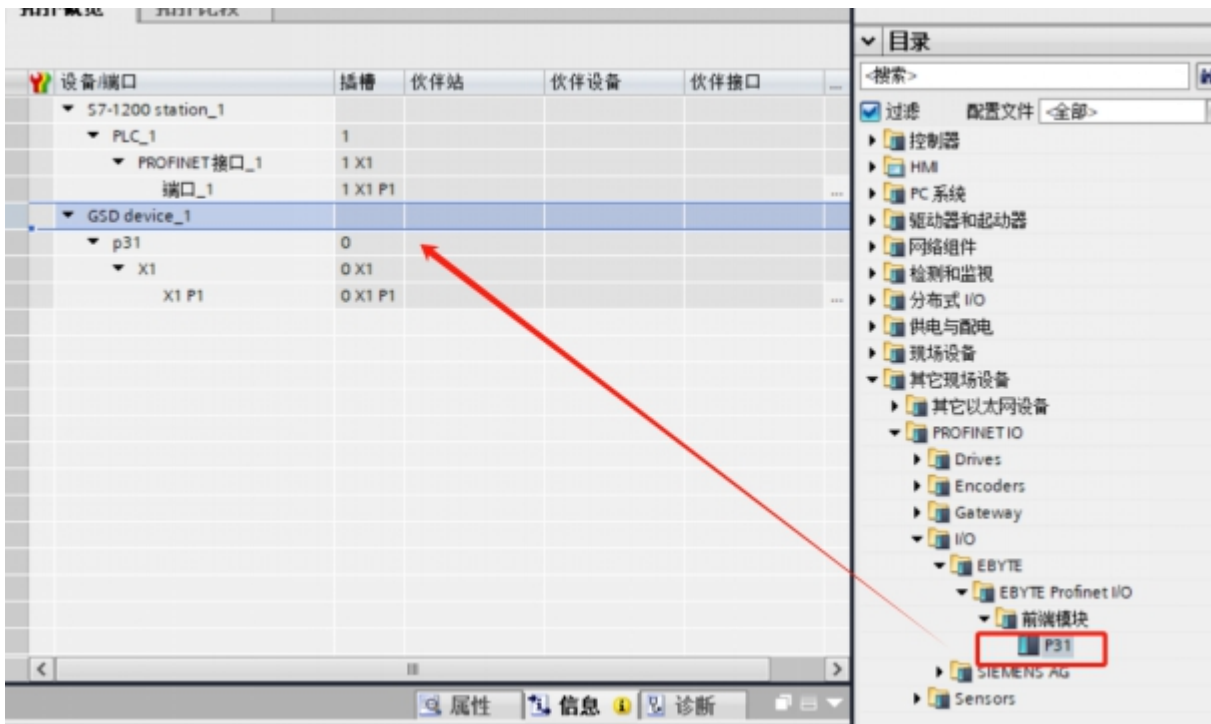
3. After the installation is complete, click Close and then you can configure the device.



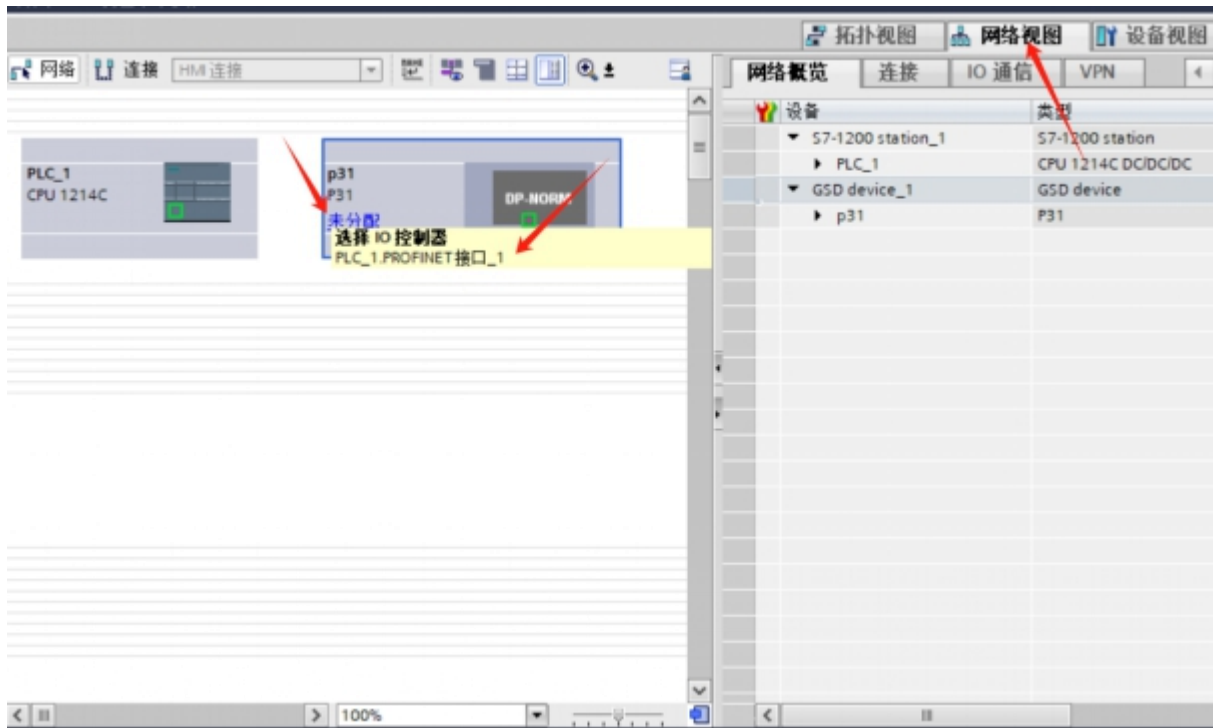
4.3 Add Profinet IO device to the project

The following example takes adding a P31-AFAX4440G-U host and a GAFAX4440-U expansion module as an example

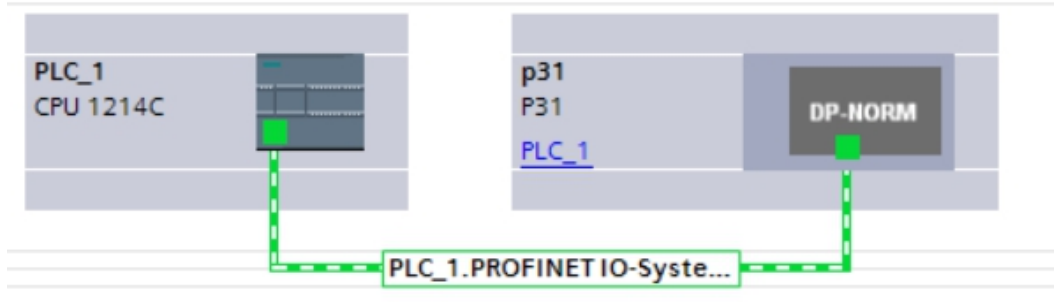
1. Create a new project or open it. If it is a new project, add the controller device first, then go to the device configuration interface, add the corresponding IO module, double-click the module, and it will be added successfully, as shown in the figure:



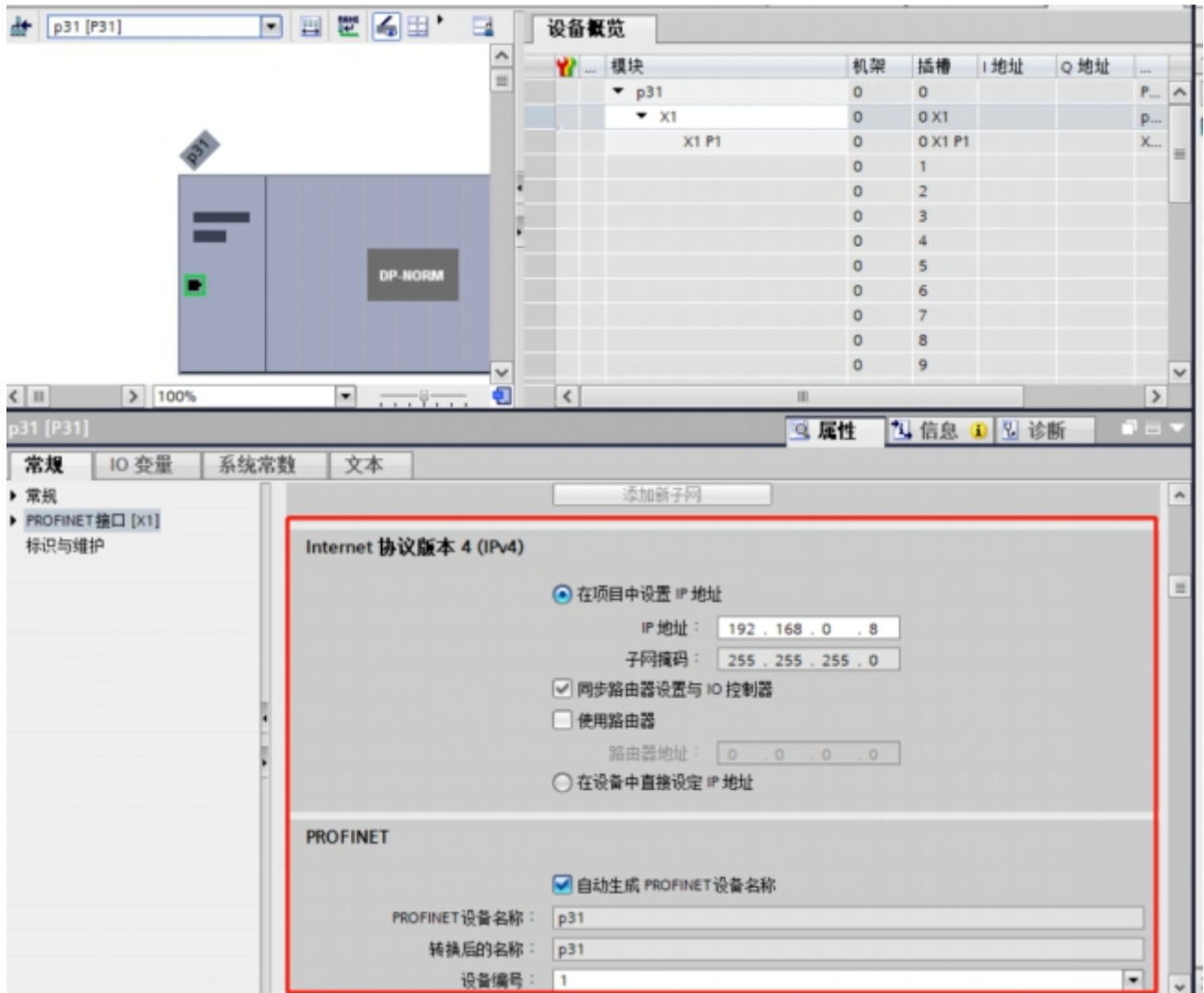
2. In the network view, configure the device



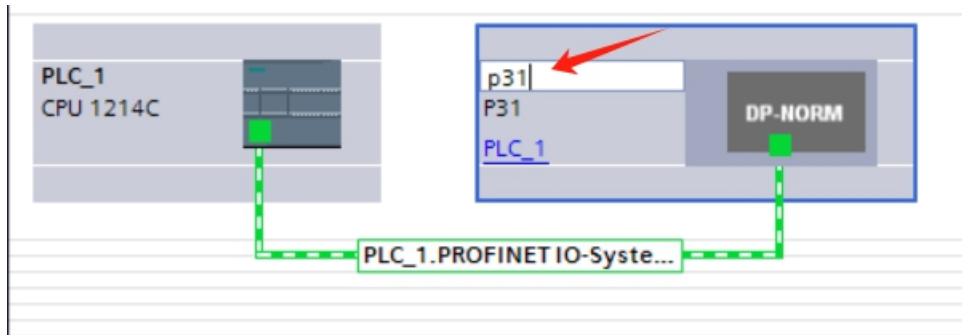
After completion, as shown in the figure:



3. In the device view, first locate and select the newly added device, then double-click the module in the picture, and then modify the IP address and device name in the general configuration interface, which must be consistent with the module itself. Alternatively, you can choose to set the IP address directly on the device.



Click here to modify the device name directly



4. Select the p31 module, click the device view, and then add P31-AFAX4440G-U and GAFAX4440-U according to the splicing order of the devices.



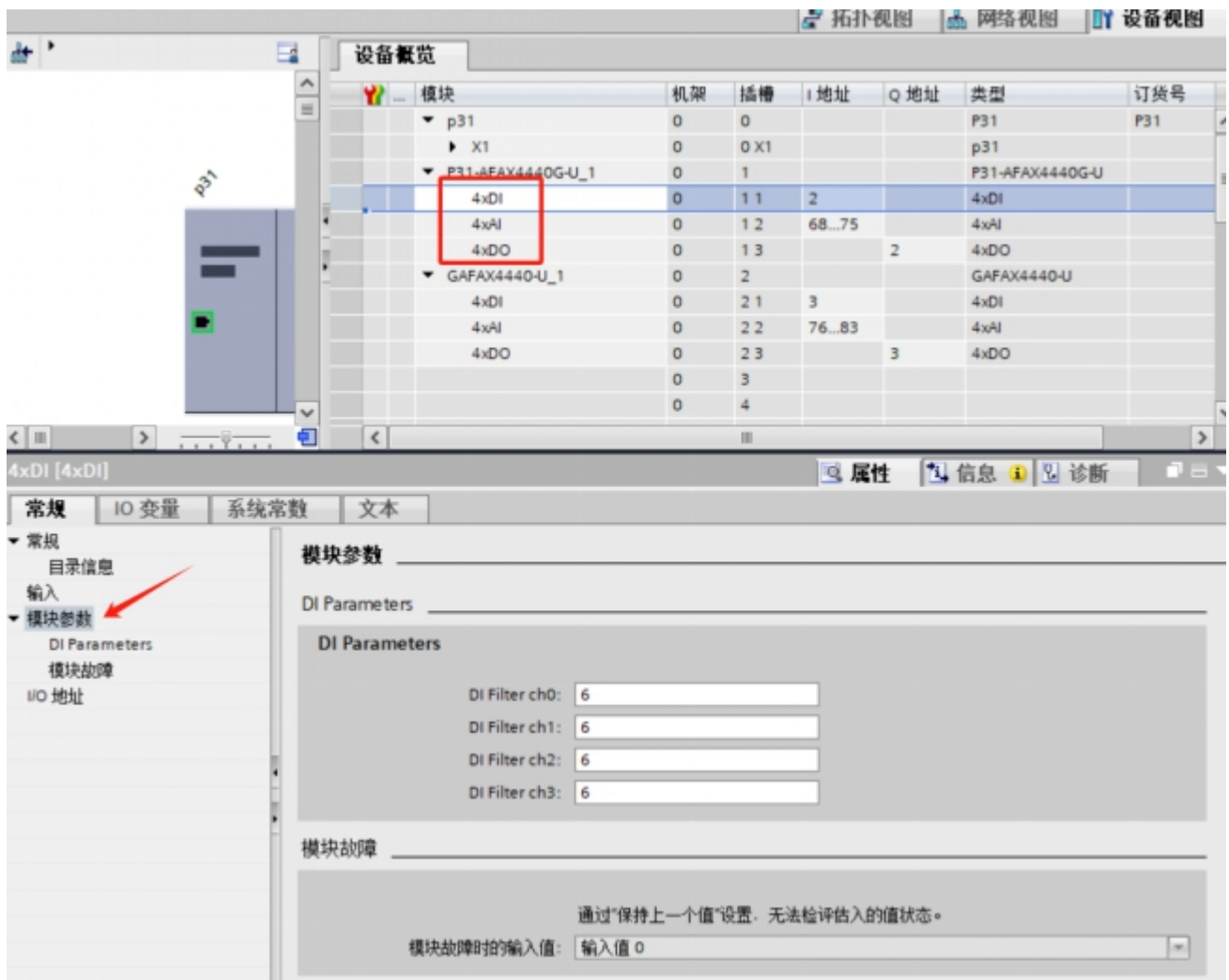
4.4 Check the starting address of I and Q points

- Through the device view of p31, you can see the address and length of each submodule:

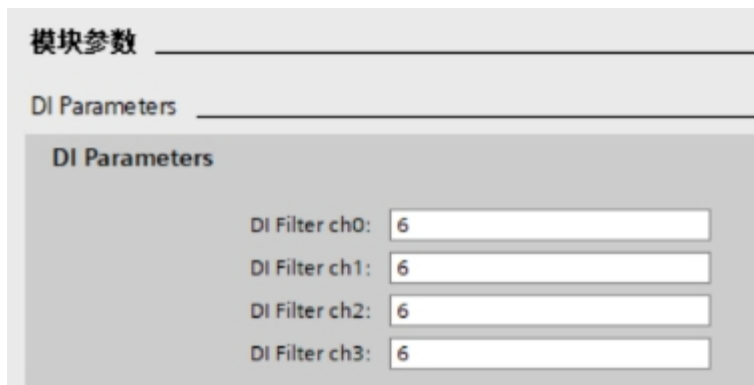


4.5 Instructions for use of IO functions:

- Click the corresponding IO type, then click the module parameters to see the relevant setting options of the IO function:



2. DI Parameters



DI Filter: You can set the DI filter time for each channel individually

3. AI Parameters

模块参数

AI Parameters

CH0

AI Filter: 0

AI zero offset: 0

AI input type: 0-20mA

CH1

AI Filter: 0

AI zero offset: 0

AI input type: 0-20mA

AI Filter: You can set the AI filter level for each channel individually. The higher the level, the longer the filtering time.

AI zero offset: (cannot be set)

AI input type: 0~20mA, 4~20mA, -20~20mA range optional

4. DO Parameters

模块参数

DO Parameters

DO Fault Output ch0: retention

DO Fault Output ch1: retention

DO Fault Output ch2: retention

DO Fault Output ch3: retention

DO Fault Output: When the bus is abnormal, the output status of DO of each channel can be set separately;

retention: When a fault occurs, the current state of the DO output is maintained.

Reset: When a fault occurs, reset the DO output to its previous state

Set: When a fault occurs, set the previous state of the DO output

5 Tutorial on loading devices using STEP 7

5.1 Preparation before connection

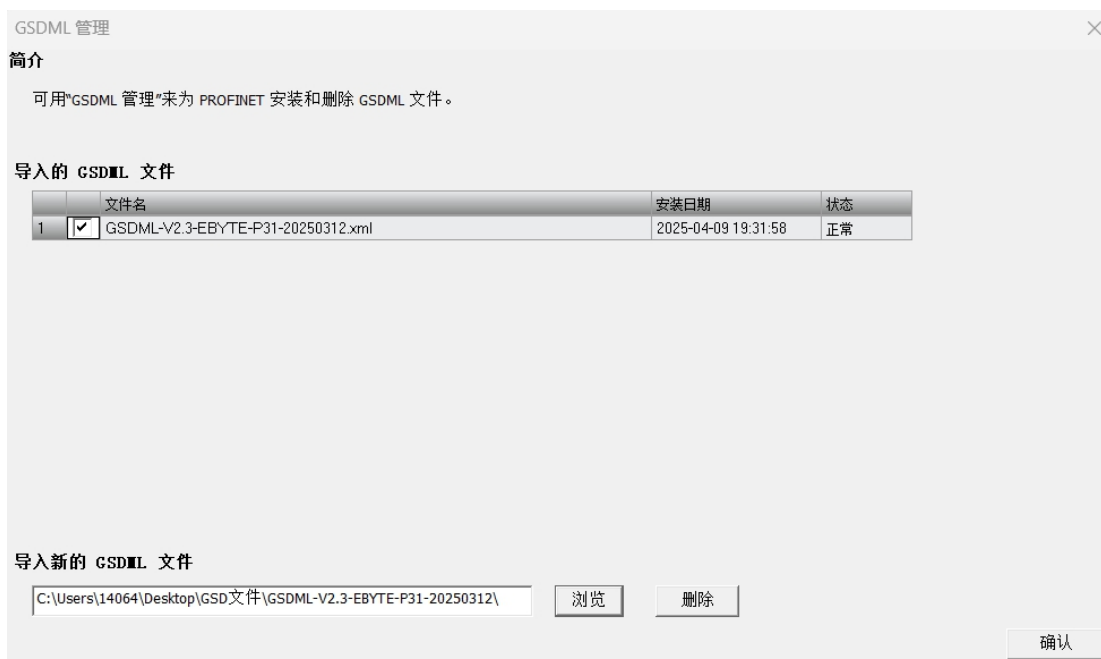
1. Prepare the required XML files as follows:
GSDML-V2.3-EBYTE-P31-20250312.xml

5.2 Add GSDML file in STEP 7

1. Under the File menu, go to GSDML Manager



2. Add GSDML file

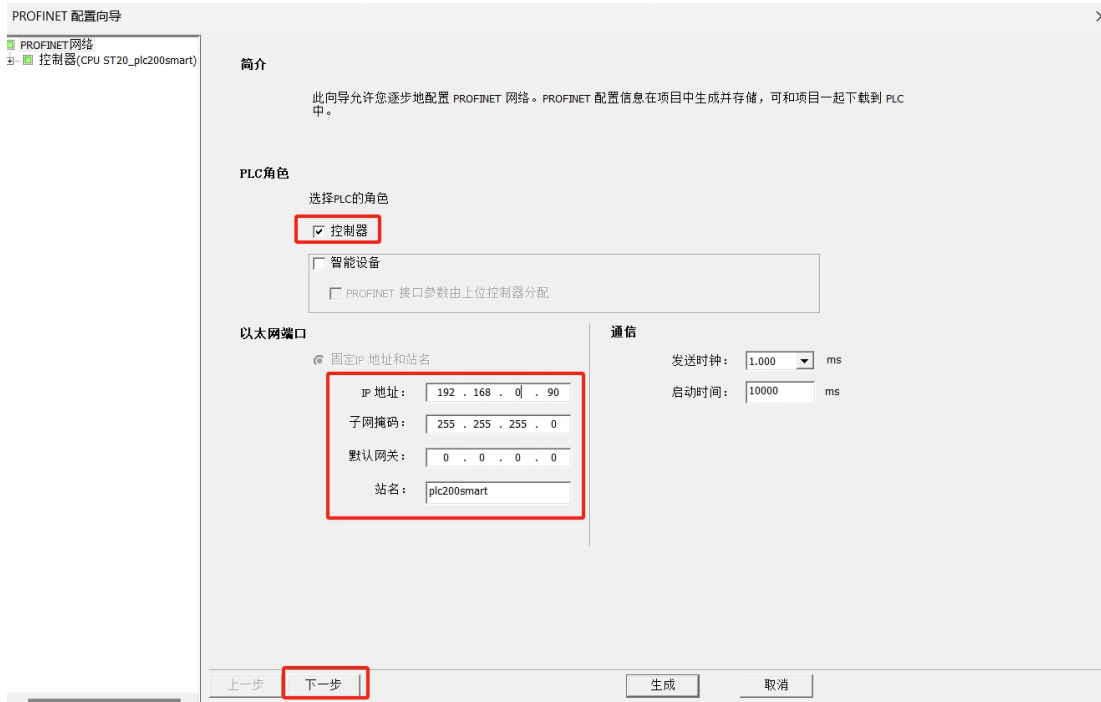


5.3 Add Profinet device to the project

1. Select the PROFINET command under the Tools menu



2. Select the PLC role as PLC controller, set the corresponding PLC controller IP and other related parameters. Click Next after completion.



3. In the right column, select PROFINET-IO→I/O→EBYTE→TION→EBYTE Profinet I/O→P31V1.0.0 by clicking on it, then hold down the left button and drag it into the table on the left.



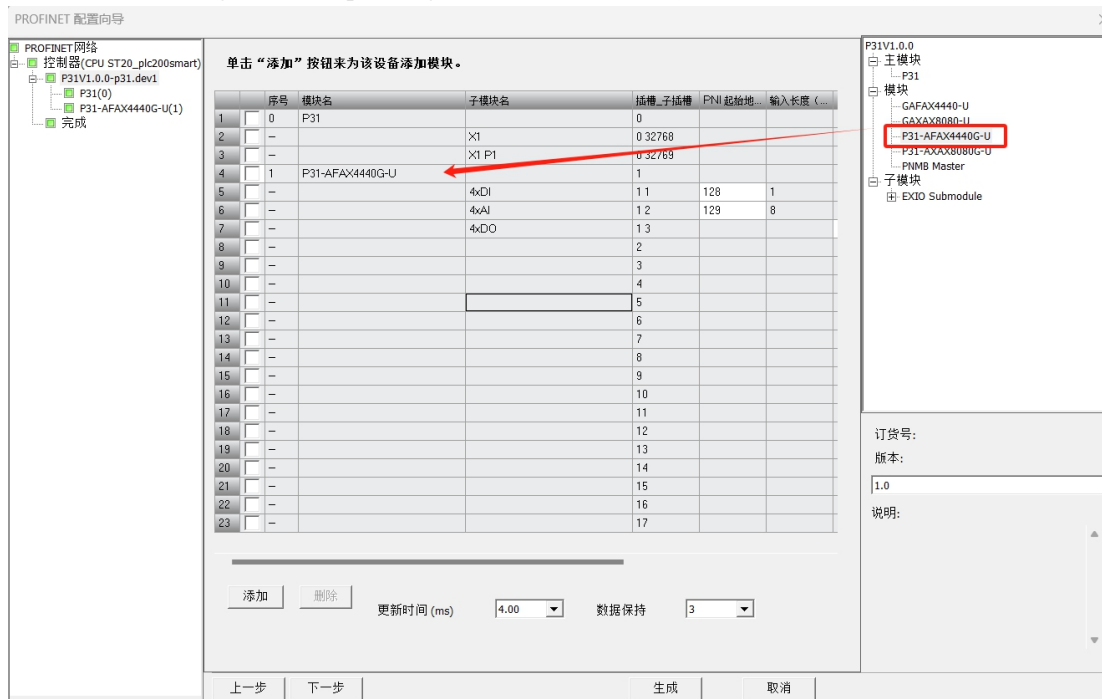
4. Double-click the device name column and fill in the corresponding device name. There cannot be the same

device name in the same project. Similarly, set the IP address and keep it in the same network segment as the PLC controller. Finally, click Generate.

Note: The device name set at this time needs to be consistent with the device. If you are not sure about the device name, you can set it randomly first, and then change the device name to be consistent. The IP address set at this time will set the IP of the device with the same device name to the value set here during configuration.



5. After completing the relevant settings, click to enter the device view operation interface. In the device overview area, drag the corresponding device model name (P31-AFAX4440G-U) into slot 1.



Note: If the P31 host has spliced IO expansion modules, you need to drag the model names of the spliced expansion modules into the slots in order. Otherwise, the device will report a configuration abnormality!

5.4 Check the starting address of I and Q points

1. Slide the bottom bar to see the starting address and length (bytes) of DI, AI, and DO.

单击“添加”按钮来为该设备添加模块。

	子模块名	插槽_子插槽	PNI 起始地址	输入长度(字节)	PNQ 起始地址	输出长度(字节)	固件版本
1		0					V1.0.0
2	X1	0 32768					
3	X1 P1	0 32769					
4		1					1.0
5	4xDI	1 1	128	1			
6	4xAI	1 2	129	8			
7	4xDO	1 3			128	1	
8		2					
9		3					
10		4					
11		5					
12		6					
13		7					
14		8					
15		9					
16		10					
17		11					
18		12					
19		13					
20		14					
21		15					
22		16					
23		17					

更新时间 (ms) 4.00 数据保持 3

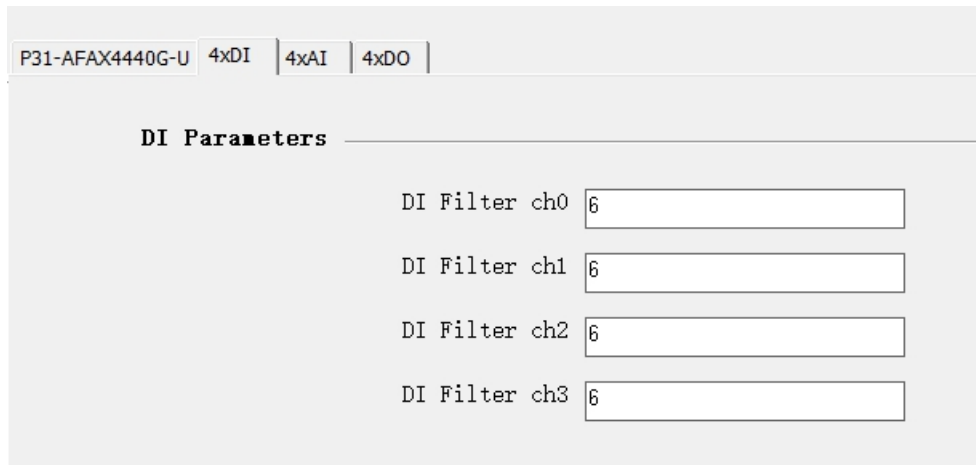
上一步 下一步 生成 取消

5.5 Instructions for use of IO functions:

1. Select the device you just added (P31-AFAX4440G-U). Here you need to configure the corresponding DI, AI, and DO related parameters according to the on-site usage requirements.



2. DI Parameters



DI Filter: You can set the DI filter time for each channel individually

3. AI Parameters

P31-AFAX4440G-U 4xDI 4xAI 4xDO

AI Parameters

CH0

AI Filter 0

AI zero offset 0

AI input type 0-20mA

CH1

AI Filter 0

AI zero offset 0

AI input type 0-20mA

CH2

AI Filter 0

AI zero offset 0

AI input type 0-20mA

AI Filter: You can set the AI filter level for each channel individually. The higher the level, the longer the filtering time.

AI zero offset: (cannot be set)

AI input type: 0~20mA, 4~20mA, -20~20mA range optional

AI input type 0-20mA

CH1

- 0-20mA
- 4-20mA
- 20mA-20mA

4. DO Parameters

P31-AFAX4440G-U 4xDI 4xAI 4xDO

DO Parameters

DO Fault Output ch0 retention

DO Fault Output ch1 retention

DO Fault Output ch2 retention

DO Fault Output ch3 retention

DO Fault Output: When the bus is abnormal, the output status of DO of each channel can be set separately;

retention: When a fault occurs, the current state of the DO output is maintained.

Reset: When a fault occurs, reset the DO output to its previous state

Set: When a fault occurs, set the previous state of the DO output

DO Parameters

DO Fault Output ch0	retention ▼
DO Fault Output ch1	retention reset set
DO Fault Output ch2	retention ▼
DO Fault Output ch3	retention ▼

After configuring the relevant parameters, click Generate.

At this moment, we have successfully completed the communication connection routine between Siemens PLC S7-200smart and Profinet distributed IO (P31-AFAX4440G-U).

6 Product Features

6.1 IO point quantity expansion

Note: When splicing equipment, do not operate with power on, otherwise it may cause damage to the equipment!

This product supports splicing 15 expansion modules, and the host can synchronize all 15 IO expansion modules in about 1ms at the fastest.

The P31-U series distributed IO host adopts an expandable structure design, in which the IO expansion module can be used for expansion with the P31-U series host. You only need to dock the IO expansion module with the host slot, and then slide the lock down to firmly connect the host and IO expansion module together.

The specific operations are as follows:

- First, make sure the host is not powered on and the host slider is at UNLK. Then connect the IO expansion module to the host, as shown below:



- After the IO expansion module is connected, slide the host slider to the LOCK position and then power on the host .
- In Portal/STEP 7, just drag the corresponding module model according to the module splicing order.

6.2 Profinet to Modbus RTU Gateway Function

1. This product provides 1 RS485 interface and supports 8 command nodes.
2. Can be used as a Modbus RTU master .
3. The Profinet to Modbus RTU gateway needs to be dragged into slot 17 before it can be used.
4. For detailed instructions on how to use the functions, please refer to the Profinet gateway manual [at https://www.ebyte.com/product/2341.html](https://www.ebyte.com/product/2341.html) .

7 Precautions

Do not splice the device while it is powered on, otherwise there is a risk of damage to the device.

Revision History

Version	Revision Date	Revision Notes	Maintainer
1.0	2025-5-6	Initial release	LT

About us

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